

Part 1: S5

1. $\Box(\Box p \rightarrow q) \vee \Box(\Box q \rightarrow p)$ in S5

1.	$\mathbb{F} 1, \Box(\Box p \rightarrow q) \vee \Box(\Box q \rightarrow p)$ ✓	Assumption
2.	$\mathbb{F} 1, \Box(\Box p \rightarrow q)$ ✓	$\vee \mathbb{F} 1$
3.	$\mathbb{F} 1, \Box(\Box q \rightarrow p)$ ✓	$\vee \mathbb{F} 1$
4.	$\mathbb{F} 2, \Box p \rightarrow q$ ✓	$\Box \mathbb{F} 2$
5.	$\mathbb{T} 2, \Box p$	$\rightarrow \mathbb{F} 4$
6.	$\mathbb{F} 2, q$	$\neg \mathbb{F} 4$
7.	$\mathbb{F} 3, \Box q \rightarrow p$ ✓	$\Box \mathbb{F} 3$
8.	$\mathbb{T} 3, \Box q$	$\rightarrow \mathbb{F} 7$
9.	$\mathbb{T} 2, q$	$\Box \mathbb{T} 8$
	⊗	

VALID

2. $\Diamond \Box p \rightarrow \Box \Diamond p$ in S5

1.	$\mathbb{F} 1, \Diamond \Box p \rightarrow \Box \Diamond p$ ✓	Assumption
2.	$\mathbb{T} 1, \Diamond \Box p$ ✓	$\rightarrow \mathbb{F} 1$
3.	$\mathbb{F} 1, \Box \Diamond p$ ✓	$\rightarrow \mathbb{F} 1$
4.	$\mathbb{T} 2, \Box p$	$\Diamond \mathbb{T} 2$
5.	$\mathbb{F} 3, \Diamond p$ ✓	$\Box \mathbb{F} 3$
6.	$\mathbb{T} 1, p$	$\Box \mathbb{T} 4$
7.	$\mathbb{F} 1, p$	$\Diamond \mathbb{F} 5$
	⊗	

VALID

3. $(\neg p \vee \neg \Box \Box p) \rightarrow \neg \Box p$ in S5

1.	$\mathbb{F} 1, (\neg p \vee \neg \Box \Box p) \rightarrow \neg \Box p$ ✓	Assumption
2.	$\mathbb{T} 1, (\neg p \vee \neg \Box \Box p)$ ✓	$\rightarrow \mathbb{F} 1$
3.	$\mathbb{F} 1, \neg \Box p$ ✓	$\rightarrow \mathbb{F} 2$
4.	$\mathbb{T} 1, \Box p$	$\neg \mathbb{F} 3$
$\swarrow$ $\mathbb{T} 1, \neg p$ ✓ $\mathbb{F} 1, p$ $\mathbb{T} 1, p$ ⊗ $\searrow$ $\mathbb{T} 1, \neg \Box \Box p$ ✓ $\mathbb{F} 1, \Box \Box p$ ✓ $\mathbb{F} 2, \Box p$ ✓ $\mathbb{F} 3, p$ $\mathbb{T} 3, p$ ⊗		
5.	$\mathbb{T} 1, \neg p$ ✓	$\vee \mathbb{T} 2$
6.	$\mathbb{F} 1, p$	$\neg \mathbb{T} 5$
7.	$\mathbb{T} 1, p$	$\Box \mathbb{T} 4$
8.	⊗	$\neg \mathbb{T} 5$
9.	$\mathbb{F} 1, \Box \Box p$ ✓	$\Box \mathbb{F} 8$
10.	$\mathbb{F} 2, \Box p$ ✓	$\Box \mathbb{F} 9$
11.	$\mathbb{F} 3, p$	$\Box \mathbb{T} 4$
	⊗	

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4. $(\Diamond p \wedge \Diamond q) \rightarrow \Box(p \vee q)$ in **K**

1.	$\mathbb{F} 1, (\Diamond p \wedge \Diamond q) \rightarrow \Box(p \vee q)$ ✓	Assumption
2.	$\mathbb{T} 1, \Diamond p \wedge \Diamond q$ ✓	$\rightarrow \mathbb{F} 1$
3.	$\mathbb{F} 1, \Box(p \vee q)$ ✓	$\rightarrow \mathbb{F} 1$
4.	$\mathbb{T} 1, \Diamond p$ ✓	$\wedge \mathbb{T} 2$
5.	$\mathbb{T} 1, \Diamond q$ ✓	$\wedge \mathbb{T} 2$
6.	$\mathbb{T} 1.1, p$	$\Diamond \mathbb{T} 4$
7.	$\mathbb{T} 1.2, q$	$\Diamond \mathbb{T} 5$
8.	$\mathbb{F} 1.3, p \vee q$ ✓	$\Box \mathbb{F} 3$
9.	$\mathbb{F} 1.3, p$	$\wedge \mathbb{F} 8$
10.	$\mathbb{F} 1.3, q$	$\vee \mathbb{F} 8$

INVALID

5. $\Diamond(p \rightarrow p) \vee \Diamond \neg(q \rightarrow q)$ in **K**

1.	$\mathbb{F} 1, \Diamond(p \rightarrow p) \vee \Diamond \neg(q \rightarrow q)$ ✓	Assumption
2.	$\mathbb{F} 1, \Diamond(p \rightarrow p)$	$\vee \mathbb{F} 1$
3.	$\mathbb{F} 1, \Diamond \neg(q \rightarrow q)$	$\vee \mathbb{F} 1$

INVALID

6. $(p \wedge \neg \neg \Box q) \rightarrow (\neg \neg p \wedge \Box q)$ in **K**

1.	$\mathbb{F} 1, (p \wedge \neg \neg \Box q) \rightarrow (\neg \neg p \wedge \Box q)$ ✓	Assumption
2.	$\mathbb{T} 1, p \wedge \neg \neg \Box q$ ✓	$\neg \mathbb{F} 1$
3.	$\mathbb{F} 1, \neg \neg p \wedge \Box q$ ✓	$\neg \mathbb{F} 1$
4.	$\mathbb{T} 1, p$	$\wedge \mathbb{T} 2$
5.	$\mathbb{T} 1, \neg \neg \Box q$ ✓	$\wedge \mathbb{T} 2$
6.	$\mathbb{F} 1, \neg \Box q$ ✓	$\neg \mathbb{T} 5$
7.	$\mathbb{T} 1, \Box q$	$\neg \mathbb{F} 6$
$\swarrow \quad \searrow$		
8.	$\mathbb{F} 1, \neg \neg p$ ✓ $\mathbb{F} 1, \Box q$	$\wedge \mathbb{F} 3$
9.	$\mathbb{T} 1, \neg p$ ✓ $\otimes$	$\neg \mathbb{T} 8$
10.	$\mathbb{F} 1, p$	$\neg \mathbb{T} 9$
	$\otimes$	

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7. $\Box(p \rightarrow q) \rightarrow (\Diamond\Diamond p \rightarrow \Diamond q)$ in S4

1.	$\mathbb{F} 1, \Box(p \rightarrow q) \rightarrow (\Diamond\Diamond p \rightarrow \Diamond q) \checkmark$	Assumption
2.	$\mathbb{T} 1, \Box(p \rightarrow q)$	$\rightarrow \mathbb{F} 1$
3.	$\mathbb{F} 1, \Diamond\Diamond p \rightarrow \Diamond q \checkmark$	$\rightarrow \mathbb{F} 1$
4.	$\mathbb{T} 1, \Diamond\Diamond p \checkmark$	$\rightarrow \mathbb{F} 3$
5.	$\mathbb{F} 1, \Diamond q$	$\rightarrow \mathbb{F} 3$
6.	$\mathbb{T} 1.1, \Diamond p \checkmark$	$\Diamond \mathbb{T} 4$
7.	$\mathbb{T} 1.1, \Box(p \rightarrow q)$	$4\Box 2$
8.	$\mathbb{F} 1.1, \Diamond q$	$4\Diamond 5$
9.	$\mathbb{T} 1.1.1, p$	$\Diamond \mathbb{T} 6$
10.	$\mathbb{T} 1.1.1, p \rightarrow q \checkmark$	$\Box \mathbb{T} 7$
11.	$\mathbb{F} 1.1.1, q$	$\Diamond \mathbb{F} 8$
$ \begin{array}{c} \swarrow \quad \searrow \\ \mathbb{F} 1.1.1, p \quad \mathbb{T} 1.1.1, q \end{array} $		
12.	$\mathbb{F} 1.1.1, p \quad \mathbb{T} 1.1.1, q$	$\rightarrow \mathbb{T} 10$
	$\otimes \quad \otimes$	

VALID

8. $(p \wedge \Diamond\Box p) \rightarrow \Box p$ in S4

1.	$\mathbb{F} 1, (p \wedge \Diamond\Box p) \rightarrow \Box p \checkmark$	Assumption
2.	$\mathbb{T} 1, p \wedge \Diamond\Box p \checkmark$	$\rightarrow \mathbb{F} 1$
3.	$\mathbb{F} 1, \Box p \checkmark$	$\rightarrow \mathbb{F} 1$
4.	$\mathbb{T} 1, p$	$\wedge \mathbb{T} 2$
5.	$\mathbb{T} 1, \Diamond\Box p \checkmark$	$\wedge \mathbb{T} 2$
6.	$\mathbb{F} 1.1, p$	$\Box \mathbb{F} 3$
7.	$\mathbb{T} 1.2, \Box p$	$\Diamond \mathbb{T} 5$
8.	$\mathbb{T} 1.2, p$	$\mathbb{T} \Box 7$

INVALID